```
clc;
clear;
close all;
```

INPUTS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

Bc = 8;  % Coefficient word length
Bp = 16; % Product word length
N = 200;
x = [1 zeros(1,N-1)]; % Unit impulse
```

1. COEFFICIENT QUANTIZATION

```
S = 2^(Bc-1)-1;
bq = round(b*S)/S;
aq = round(a*S)/S;

CoeffError = [b-bq a-aq];
```

2. IDEAL AND COEFFICIENT-QUANTIZED RESPONSE

```
y_ideal = filter(b,a,x);
y_coeff = filter(bq,aq,x);

e_coeff = y_ideal-y_coeff;
MaxCoeffError = max(abs(e_coeff));
SQNR_coeff = 10*log10(sum(y_ideal.^2)/sum(e_coeff.^2));
```

3. PRODUCT QUANTIZATION

```
Sp = 2^(Bp-1)-1;
y_prod = zeros(1,N);
v = zeros(1,4);

for n = 1:N
    p = bq*x(n);
    fb = -aq(2:5).*v;
    p = round([p fb]*Sp)/Sp; % Product quantization
    y_prod(n) = sum(p);
    v = [y_prod(n) v(1:3)];
end

e_prod = y_coeff-y_prod;
```

```
MaxProdError = max(abs(e_prod));
SQNR_prod = 10*log10(sum(y_coeff.^2)/sum(e_prod.^2));
```

4. ZERO-INPUT LIMIT-CYCLE TEST

```
x0 = zeros(1,N);
state = [0.01 0.01 0.01 0.01];
y_zero = zeros(1,N);

for n = 1:N
    acc = -aq(2:5)*state.';
    y_zero(n) = round(acc*Sp)/Sp;
    state = [y_zero(n) state(1:3)];
end

LimitAmplitude = max(abs(y_zero));
LimitSamples = find(abs(y_zero)>1e-12);
```

5. OVERFLOW / SATURATION TEST

```
xo = 1.2*ones(1,N);
y_over = filter(bq,aq,xo);

% Normalized saturation
y_sat = min(max(y_over,-1),1);

OverflowCount = sum(abs(y_over)>1);
MaxOverflow = max(abs(y_over));
```

6. OUTPUT TABLE

```
fprintf('\n--- QUANTIZATION RESULTS ---\n');
fprintf('Maximum coefficient/output error = %.6e\n',MaxCoeffError);
fprintf('Coefficient SQNR = %.2f dB\n',SQNR_coeff);
fprintf('Maximum product quantization error = %.6e\n',MaxProdError);
fprintf('Product SQNR = %.2f dB\n',SQNR_prod);

fprintf('Zero-input maximum amplitude = %.6e\n',LimitAmplitude);
fprintf('Overflow samples = %d\n',OverflowCount);

T = table( ...
    ["Coefficient Quantization";
     "Product Quantization";
     "Zero-Input Test";
     "Overflow Test"], ...
    [MaxCoeffError; MaxProdError; LimitAmplitude; MaxOverflow], ...
    [SQNR_coeff; SQNR_prod; NaN; NaN], ...
    'VariableNames', ...
    {'Test','Maximum_Error_or_Amplitude','SQNR_dB'});

disp(T);
```

```
--- QUANTIZATION RESULTS ---
Maximum coefficient/output error = 4.405764e-03
Coefficient SQNR = 32.23 dB
Maximum product quantization error = 1.651951e-01
Product SQNR = 1.57 dB
Zero-input maximum amplitude = 9.216590e-03
Overflow samples = 194
```

Test	Maximum_Error_or_Amplitude	SQNR_dB
"Coefficient Quantization"	0.0044058	32.233
"Product Quantization"	0.1652	1.5673
"Zero-Input Test"	0.0092166	NaN
"Overflow Test"	1.3599	NaN

7. QUANTIZED COEFFICIENTS

```
disp('Original b =');
disp(b);

disp('Quantized b =');
disp(bq);

disp('Original a =');
disp(a);

disp('Quantized a =');
disp(aq);
```

```
Original b =
    0.0048    0.0193    0.0289    0.0193    0.0048

Quantized b =
    0.0079    0.0157    0.0315    0.0157    0.0079

Original a =
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Quantized a =
    1.0000   -2.3701    2.3150   -1.0551    0.1890
```

8. PLOTS

```
figure;
stem(0:N-1,y_ideal,'filled');
hold on;
stem(0:N-1,y_coeff);
grid on;
xlabel('n');
ylabel('Amplitude');
```

```
title('Ideal vs Coefficient-Quantized Response');
legend('Ideal','Quantized');

figure;
plot(0:N-1,e_coeff,'LineWidth',1);
grid on;
xlabel('n');
ylabel('Error');
title('Coefficient Quantization Noise');

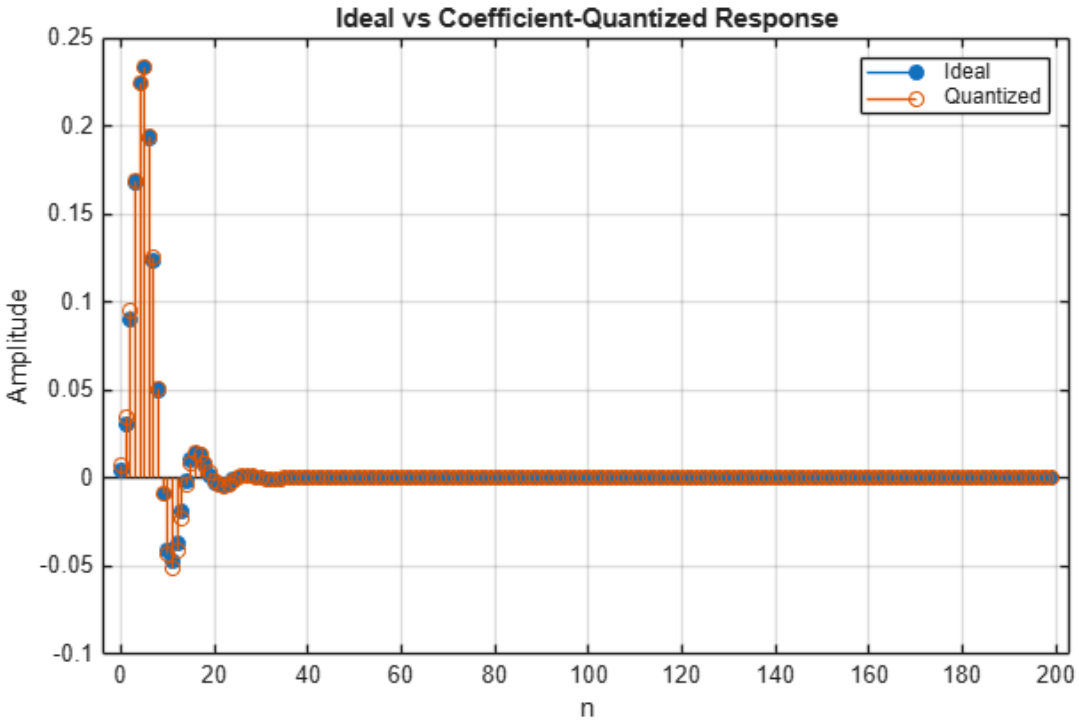
figure;
stem(0:N-1,y_coeff-y_prod,'filled');
grid on;
xlabel('n');
ylabel('Error');
title('Product Quantization Noise');

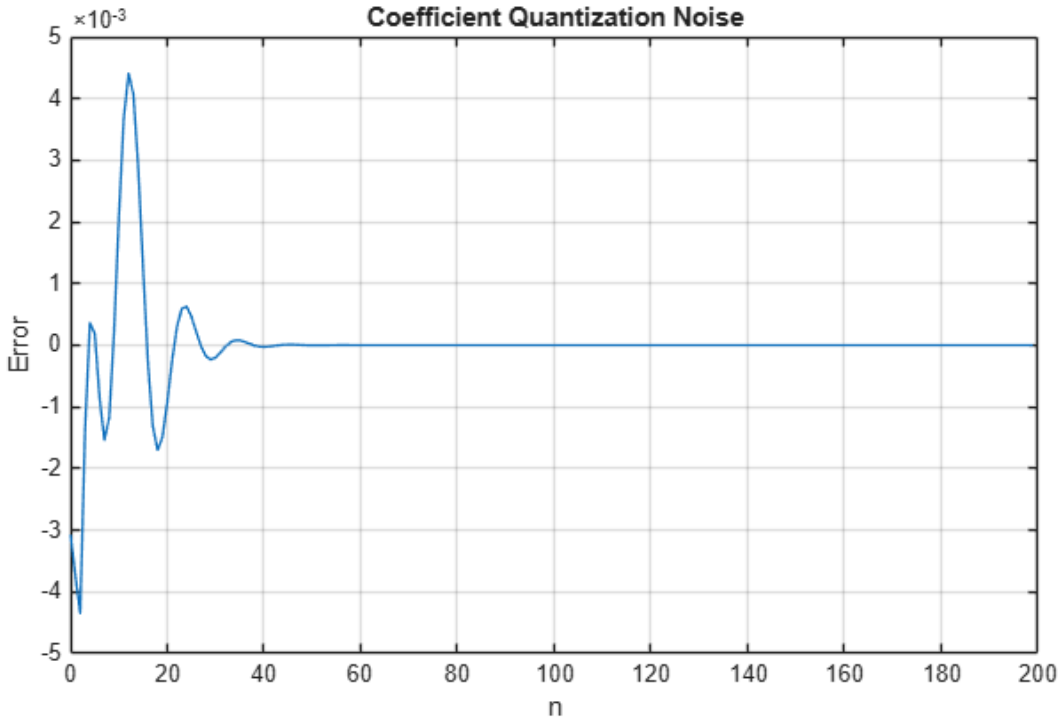
figure;
plot(0:N-1,y_zero,'LineWidth',1);
grid on;
xlabel('n');
ylabel('Amplitude');
title('Zero-Input Limit-Cycle Test');

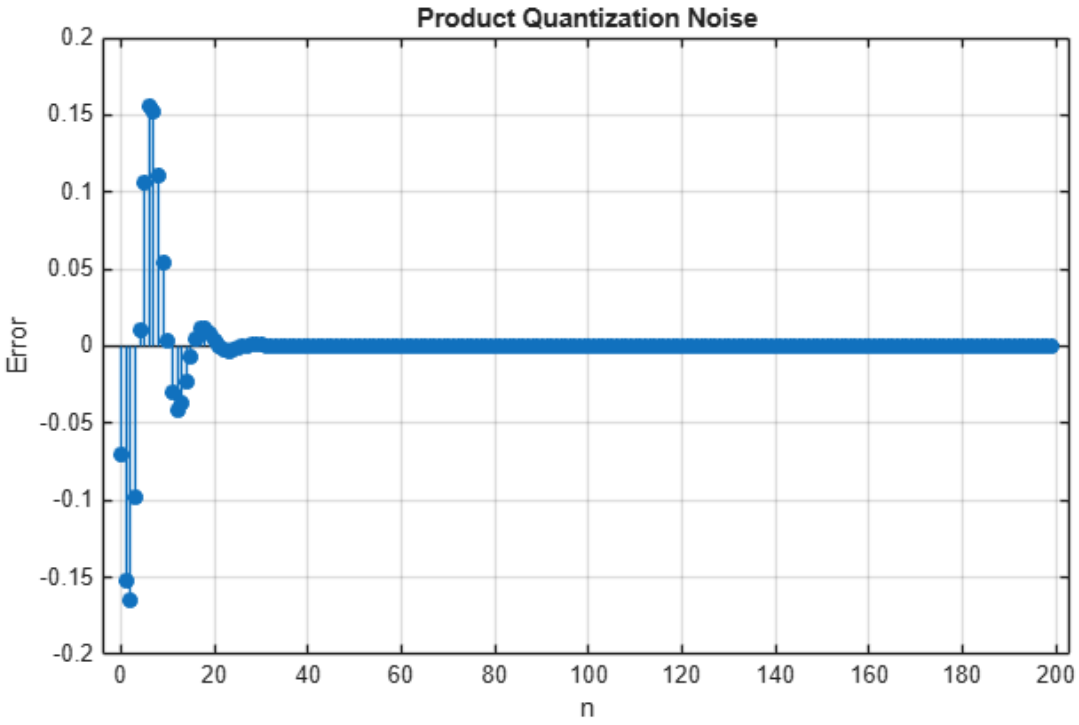
figure;
plot(0:N-1,y_over);
hold on;
plot(0:N-1,y_sat,'--');
grid on;
xlabel('n');
ylabel('Amplitude');
title('Overflow / Saturation Test');
legend('Without Saturation','With Saturation');

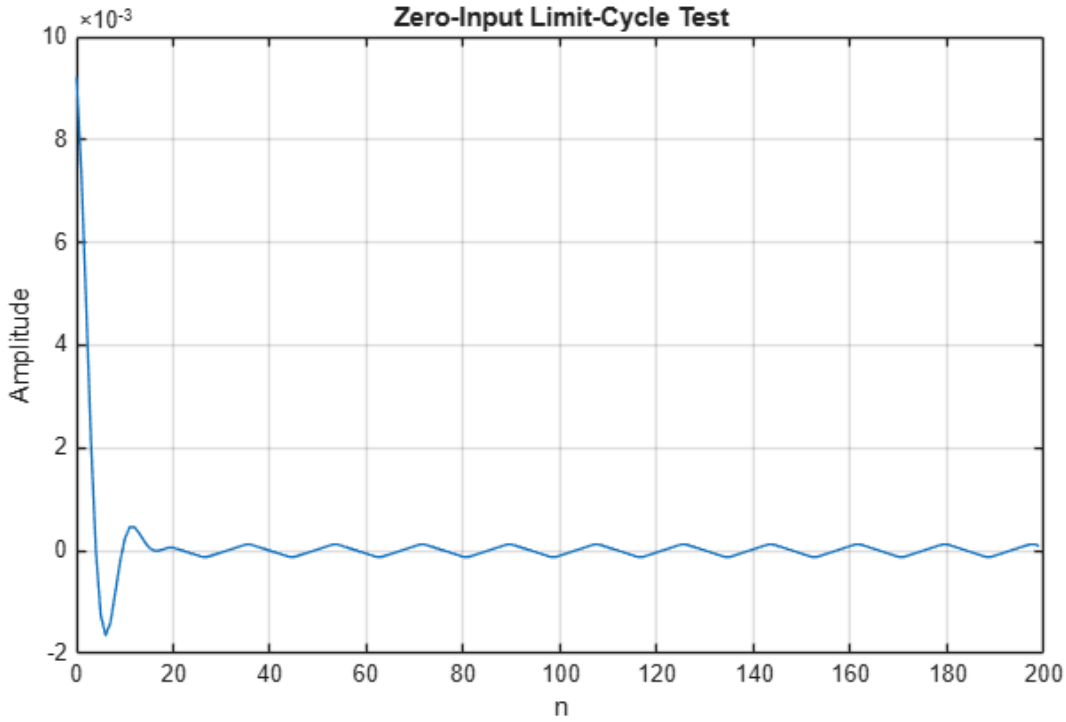
figure;
zplane(b,a);
grid on;
title('Ideal Filter Pole-Zero Plot');

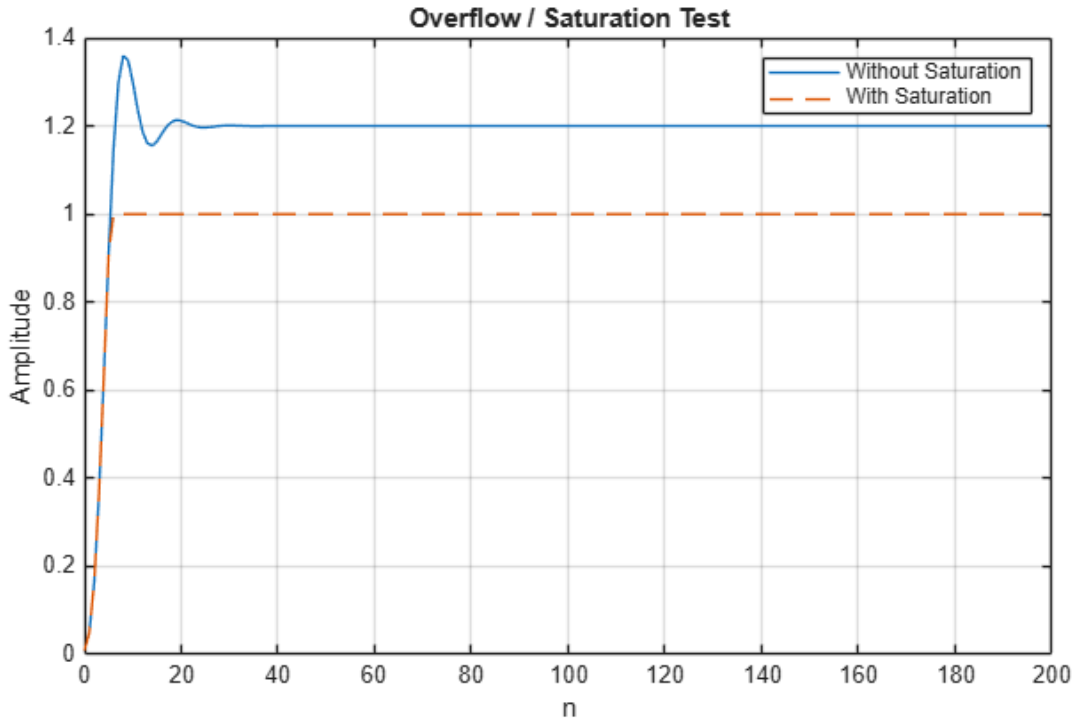
figure;
zplane(bq,aq);
grid on;
title('Quantized Filter Pole-Zero Plot');
```

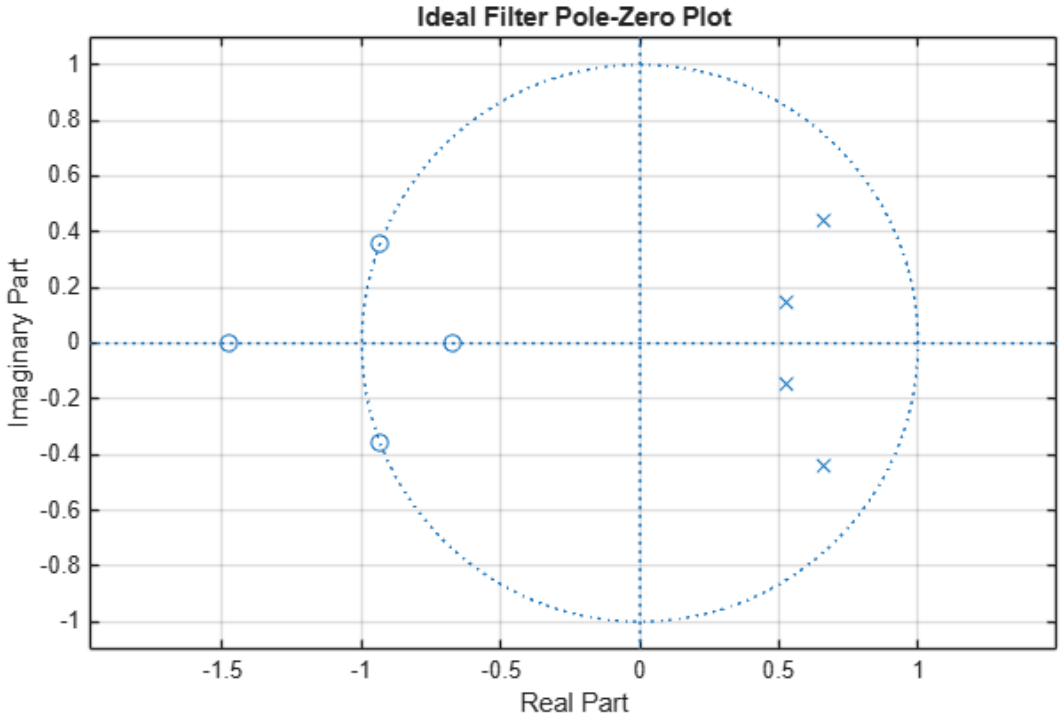

Ideal vs Coefficient-Quantized Response
Ideal
Quantized
Amplitude
n
0.25
0.2
0.15
0.1
0.05
0
-0.05
-0.1
0
20
40
60
80
100
120
140
160
180
200

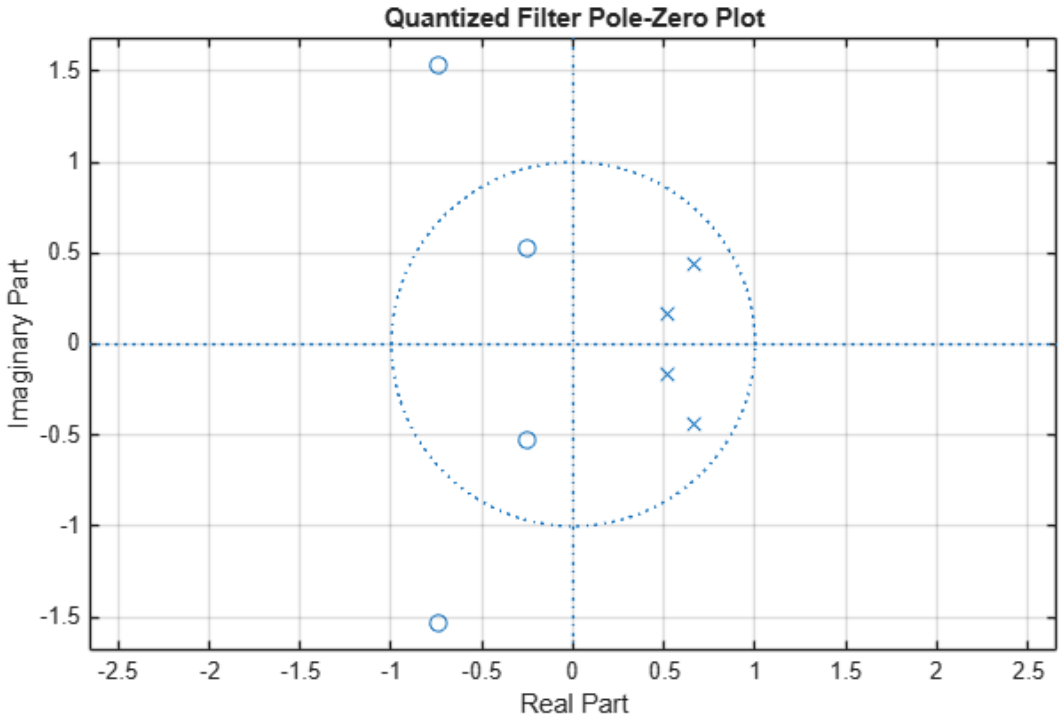

Coefficient Quantization Noise
×10-3
Error
n
5
4
3
2
1
0
-1
-2
-3
-4
-5
0
20
40
60
80
100
120
140
160
180
200


Product Quantization Noise
Error
n
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.2
0
20
40
60
80
100
120
140
160
180
200


Zero-Input Limit-Cycle Test
×10-3
Amplitude
n
10
8
6
4
2
0
-2
0
20
40
60
80
100
120
140
160
180
200


Overflow / Saturation Test
Without Saturation
With Saturation
Amplitude
n
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
20
40
60
80
100
120
140
160
180
200


Ideal Filter Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1



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